

Sample Exam Questions & Reference Guide

Question 1: Algorithm Analysis

Question: What is the worst-case time complexity of searching for an element in an unbalanced Binary Search Tree?

Reference Answer: $O(n)$ worst-case complexity occurs when the BST degenerates into a single linear linked list.

Question 2: Memory Management

Question: Differentiate between Array and Linked List memory allocation.

Reference Answer: Arrays allocate contiguous memory blocks, enabling fast index access. Linked lists allocate non-contiguous memory dynamically using pointers.